DEPARTMENT OF COMPUTING AND INFORMATICS

THESIS AND CAPSTONE GUIDELINES

A. COURSE DESCRIPTION

CAPSTONE-IT	THESIS-CS
Course Code: CAP 302 **Units:** 3 units **Course Title:** Capstone Project1 **Prerequisite:** Regular 3rd Year **Course Description:** This course allows the student to formulate the project objectives, scope and limitations, and evaluation metric; compare and collect literature related to the project; and propose an ethical and feasible IT solution to the identified problems in the project. **Requirement:** Chapter 1-2-3	**Course Code:** CS 302 **Units:** 3 units **Course Title:** Thesis 1 Pre-requisite: Regular 3rd Year **Course Description:** This course allows the student to formulate the project objectives, scope and limitations, and evaluation metric; compare and collect literature related to the project; and propose an ethical and feasible IT solution to the identified problems in the project. **Requirement:** Chapter 1-2-3
Course Code: CAP 402 **Units:** 3 units **Course Title:** Capstone Project 2 Pre-requisite: Regular 4th Year **Course Description:** This course allows the student to formulate the project objectives, scope and limitations, and evaluation metric; compare and collect literature related to the project; and propose an ethical and feasible IT solution to the identified problems in the project. **Requirement:** Chapter 4-5	**Course Code:** CS 401 **Units:** 3 units **Course Title:** Thesis 2 **Pre-requisite:** Regular 4th Year **Course Description:** Collect pertinent data to support research objectives of the thesis. Design the architecture and components of the proposed software solution. Justify the proposed solution's feasibilities and effectiveness to solve the computing problem. **Requirement:** Chapter 4-5

B. **CONDUCT FOR DEFENSE**

TOPIC DEFENSE

1. Each group must submit three(3) copies of the concept paper signed by their advisers at least one week /(7) days, until 3:00 pm before the Defense Date.
2. The group who fails to submit their concept paper on time will receive deductions.
3. Each group will be given 30 minutes to present and defend their titles.

The breakdown is provided below:

Title Defense
15 minutes – presentation 15 minutes – clarifications (Q and A)

PROPOSAL AND FINAL DEFENSE

4. Each group must submit six(6) copies (1 soft copies and 5 hard copies) of their research paper together with the adviser's recommendation (attachment of recommendation form) at least one week /(7) days, until 3:00 pm before the Defense Date.
5. The group who fails to submit their research paper on time will fall on the redefense category. Ceiling grade for redefense is 3.00.
6. Each group will be given one hour to present and defend their research paper.

The breakdown is provided below:

Proposal Defense	**Final Defense**
10 minutes - presentation of manuscript (Chapter 1-3) 20 minutes - presentation of system 20 minutes – Question and Answer 10 minutes - Deliberation	10 minutes (3 minutes summary of chapter1 -3 and 7 minutes 4-5) - presentation of manuscript (Chapter 1-5). 20 minutes - presentation of system 20 minutes – Question and Answer 10 minutes - Deliberation

7. The group are expected to be in their business attire during the research paper presentation.

C. **Grading System**

Criteria for Grading

Topic Defense	Proposal Defense	Final Defense
The Researchers will present three(3) titles. The panel will decide on which particular title should be pursue.	Oral Presentation 20% Ability to Answer Questions 30% Feasibility of Research 50% Grade Total 100%	Oral Presentation 20% Ability to Answer Questions 30% Software Quality 50% • Usability (10%) • Functionality (10%) • Efficiency (10%) • Reliability (10%) • other factors (10%) Grade Total 100%

GRADESHEET FOR PRE-ORAL DEFENSE (Thesis/Capstone)

Office of the Director of
Department of Computer Education

THESIS 1 GRADING SHEET
PRE-ORAL DEFENSE
2nd Semester, F.Y. 2020-2021

Group No.: _____ Date: ____
Title : _____

The rating of each proponent per panel member shall be based on the following rubric for objective evaluation purposes:

Pre- Oral Defense

Student's Name	Q & A (30%) – Individual Grade	Presentation (20%) - Group Grade	Content of Research Proposal (50%) – Group Grade	GRADE

Comments and Suggestions:
_____
_____
_____
_____

VERDICT: _____

There will be four possible verdicts after the Proposal. The verdict is a unanimous decision among the three members of the Oral Defense panel. Once issued, it is final and irrevocable.

APPROVED. Minor revisions are necessary but they do not have to be presented in front of and checked by all panelists. 91 – 100
APPROVED WITH MINOR REVISIONS. Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. 86 – 90
APPROVED WITH MAJOR REVISIONS. Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. 60 – 85
DISAPPROVED / REDEFENSE. The Proponents/Researchers failed to propose a researchable or scholarly Research / Capstone Project. Below 60

NAME OF EVALUATOR: ____
Date_____

GRADESHEET FOR ORAL DEFENSE (Thesis/Capstone)

Office of the Director of
Department of Computer Education

THESIS 1 GRADING SHEET
ORAL DEFENSE
2nd Semester, F.Y. 2022-2023

Group No.: _____ Date: ____
Title : _____

The rating of each proponent per panel member shall be based on the following rubric for objective evaluation purposes:

Oral Defense

Student's Name	Q & A (30%) – Individual Grade	Presentation (20%) - Group Grade	Content of Research Proposal (50%) – Group Grade	GRADE

Comments and Suggestions:
_____
_____
_____
_____

VERDICT: _____

There will be four possible verdicts after the Proposal. The verdict is a unanimous decision among the three members of the Oral Defense panel. Once issued, it is final and irrevocable.

APPROVED. Minor revisions are necessary but they do not have to be presented in front of and checked by all panelists. 91 – 100
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APPROVED WITH MAJOR REVISIONS. Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. 60 – 85
DISAPPROVED / REDEFENSE. The Proponents/Researchers failed to propose a researchable or scholarly Research / Capstone Project. Below 60

NAME OF EVALUATOR: ____
Date_____

D. Organization of the Manuscript

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- **Research Framework**
  - ***Theoretical Framework***
  - ***Conceptual Framework***
- **Research Objectives**
  - ***General Aim***
  - ***Specific Objectives***
- **Significance of the Study**
- **Scope and Limitation**
- **Definition of Terms**

Chapter 2 LITERATURE REVIEW

- ***Synthesis***

Chapter 3 METHODOLOGY

- **Research Design**
- **Research Locale**
- **Population of the Study**

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Instruments of the Study

Data Analysis

Software Development Methodology

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Chapter 1

THE PROBLEM AND ITS BACKGROUND

(no introductory paragraph)

This section gives the reader the rationale of researcher's work and provides a sufficient background, relevance, and importance of the study. It will give an overview of the topic's condition in the current international and local settings. The proponents must further explain why there is a need for the software to be developed and it should be in conformity to the gap in the technology that the proponents found during the gap analysis. Included also are the algorithms to be used in solving the gap in the technology that would serve as the research.

Project Context

It contains the general problem of the study that the proposed system will address. It also enumerates the specific problems that will be solved once the proposed system is completed.

Research Objectives

This section outlines the specific, measurable goals of the study, providing a clear roadmap for investigating and addressing the research aims, ultimately contributing to the existing body of knowledge in the field.

General Aim

This section provides what the study intends to achieve by providing a concrete output of the study and how it will be achieved.

Specific Objectives

This section contains the specific steps that the researcher must undertake to meet the general objective. At least five must be included in the study, including the 2 required objectives, i.e. proving of algorithm, and ISO standards.

Research Framework

This section discusses the important theory/theories used by the researcher in the course of designing or developing the thesis. The theory/theories serve as the scientific anchor or support in conducting the study and should be derived from the Review of Literatures.

Theoretical Framework

This section discusses the theories that explore the interpretations that people make of the present issues which also outlines the appropriate philosophical assumptions of the study. Thus, this could be discussed in two different ways: nature and use in the study, and its philosophical assumptions (Cresswell, 1998). The discussion of the underpinning theories shall delve on how the theory shall be used in the analysis of the collected data.

(If underpinning theories do not have a role in the analysis or in the variables under consideration, then do not include theoretical framework)

Conceptual Framework

This is a structured set of concepts, assumptions, expectations, beliefs, and theories which main purpose is to underpin and guide research where it can be effectively conveyed through visual or graphical representations (Maxwell, 2005). It is the diagrammatic figure on the role of the variables among each other and how the study's process will undergo.

Figure 1-1. IPO Chart

Significance of the Study

This section must explain why the research focuses in this area, the expected contributions and benefits from the research. The beneficiaries should be stated. The researchers must arrange it to whom will benefit the most to the least beneficiary.

Scope and Limitation

This section includes first, the coverage of the research including but not limited to research design, theory, sampling and participants, and data analysis. Second, are the functionalities that the system will and will not perform and the limitations that the researchers right away declared based on the capacities they can perform. Last, these are constraints that the researchers wished to do but stop them from doing because of several circumstances that are beyond their control.

Definition of Terms

This section includes written description of the terms or concepts which are relevant to the research. It must contain conceptual and operational key terms which must be defined in both theoretical with cited source and connotational definitions based on present study's context.

Chapter 2

LITERATURE REVIEW

This section talks about the features, capabilities, and limitations of existing researches, algorithms, or software that are relevant and closely related/similar to the proposed topic. The proponents will present their review of literatures in thematic order following the inverted pyramid (global, local, and focal) context.

Synthesis

First, the researchers cite authors with similar results and how these similarities helped the present study. Second, the researchers cite authors with different results and how these differences helped the present study. Last, the researchers discuss where do the previous study fall short and how the present study will address them (Research Gap).

Chapter 3

METHODOLOGY

This section discusses the systematic approach to the analysis and design of the system. The research may use either structured or an object-oriented approach of the system development.

Research Design

This section states what research methodology will be utilize (quantitative, qualitative, or Mixed Methods) and from the method what specific research design will be used by describing the design based from a source. Justify why this design is the most appropriate design that explains the research process.

Research Locale

Describe physically the location of the chosen research site. The researchers may (optional) include the historical background, vision and mission, and how these details justify the appropriateness of the site why it's indeed chosen.

Population and Participants of the Study

This section discusses the nature, who and what characteristics the population under consideration have and from this population, who will be the samples. Then, explain the reason why these samples are appropriate to the present study.

Sampling Technique

This section discusses the sampling technique that the researcher will use to acquire the samples from the population. The researchers may present the computation on how the minimum sample size will be derived and they may also present the population and sample distribution table.

Inclusion and Exclusion Criteria

If the samples taken are not randomize, provide the detailed included and excluded criteria on how they choose the participants. If randomize, this section shall be omitted.

Instruments of the Study

This section will discuss the tools the researcher will use to collect the necessary data and information. It includes the brief description of the contents and how the instruments undergo validity and reliability.

Data Analysis

This section discusses the enumerated analyses utilized i.e statistical treatment (for quantitative data) including the brief description of the variables, formula, description of the symbols in the formula, and table for interpretation, or thematic coding for example (for qualitative data) including cited literature on what the utilized analysis is all about and the detailed procedure which the researchers made i.e from transcription to axial and open coding down to categories until emerging themes were derived, or both (for mixed data). Tools should be chronologically arranged based on the organization posted in the chapter 1 under the research objectives section.

Software Development Methodology

This section provides an overview of the software methodology, detailing the system, software development process, project schedule, resource allocation, algorithm description, and its practical application.

System Overview

This section gives an overview of the main features and capabilities of the system.

Software Development Process

This discusses the software methodology model the researcher will use and follow. It will also discuss the different activities that will be performed in every phase.

It will illustrates and shows all activities, persons involved, and duration of the activities (date started to date finished), activity dependencies, and activity milestone or deliverables as depicted using Gantt Chart (Activity Chart).

Systems Model (i.e, DFD, Flowcharts, HIPO, ERD, Screen Design, Algorithm Description and Application of Algorithm.

Inclusion of other system model (i.e, Architectural Design, Decision Tree, Use Case) depending on the needs of the study.

Ethical Consideration

This section discusses the enumerated ethical issues such as respect beneficence and justice and how the researchers address these issues. The researchers will explain the purpose, benefits, voluntary participation, anonymity, truth, accuracy, privacy and security from the potential harm of the study

Chapter 4

RESULTS AND DISCUSSION

This section presents the chronological arrangement of the research questions in statement form posted in chapter 1 anchored in the results of the data analysis and interpretation of the data including justifications based on literatures.

1. **Objective 1**

   Brief introduction of the table.

   Table 4-1. Name of the table name of the table name of the table name of the table name of the table name of the table name of the table

Chapter 5

SUMMARY, CONCLUSION, AND RECOMMENDATION

This section discusses the most significant findings, interrelations of all the variables of the study, and specific recommendations for further research or practical steps that can be taken.

Summary

Summary of the Study

This section recalls the reader about the entire study. It is the reiteration of the theory, research objectives, research design, participants, and data analysis.

Summary of Results

This section provides a concise and clear presentation of the main findings based on the chronological arrangement of the research questions posted in Chapter 1. This must present the itemized empirical results (i.e general weighted mean values, p-values, or major themes). This shall highlight the strengths and weaknesses found in the study.

Conclusion

This section presents the summary of the key findings, discuss their implications, and provide the interrelations of all the findings in each research questions written in paragraph form.

Recommendation

This section presents the researchers' enumerated suggestions of practical steps, policy changes, or future research directions that stem from the results gained in the study and should contribute to addressing the objectives raised in the research. Hence, it should have alignment from the research questions, results and findings, and its corresponding recommendation.

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REFERENCES

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D. Formatting Guidelines (IEEE)

Citation and Reference Style 1. References in the Text

- References must be numbered in the same order as they occur in the text.
- Use the same number in all subsequent references after source has been identified.
- The text line should have square brackets around each reference number, with a space between the bracket and the punctuation: "... end of the line for my research [12]."
- Unless it is pertinent to the paper, it is not required to indicate the author(s) of the reference. Never include the reference's date in the text.
- "In [27]..." is adequate and does not require the phrase "in reference [27]".

• To reference several sources:

o [1],[3],[5] o [1-10]

[1,3,7] o [1-10]

• Except in cases when there are more than six writers, all names of authors should be listed in the references. Use et al. (no comma before et) after the name of the first author if there are more than six writers.

2. The Reference List

- References must be provided in numerical sequence, according to how they were referenced. There must be no alphabetical arrangement for the references.
- Each entry's lines should be indented, and the number in brackets should be on the line.
- Provide only one reference for each number in brackets.

3. Books

Author initials. Last name, *Book Title*, City (and state if in US), Country: Publisher, Year.

4. Periodicals

[2] A. A. Author, "Name of paper," *Abbreviated Title of Periodical*, vol. x, no. x, pp. xxx-xxx, Abbrev. Month, year, doi: xxx

5. Articles from Published Conference Proceedings

The word *in* before the conference title should not be italicized.

[3] Author and B. Author of paper, "Title of paper," in Title of Published Proceedings: Proceedings of the Title of Conf.: Subtitle of conference, Month Day, Year, Location, [Format]. Available from Database Name (if appropriate), doi:.

6. Papers Presented at Conferences, but Unpublished

[4] Author and B. Author of paper, "Title of paper," presented at Title of Conf.: Subtitle of conference, Month Day, Year, Location, [Format].

7. Reports (technical reports, internal reports, memoranda)

Provide number and month if available.

[5] Author(s) First name or initials. Surname, "Title of report," Abbrev. Name of Co. or Inst., City of Co., Abbrev. State, Rep. number, Year of Publication. Available from Database Name (if appropriate)

8. Thesis or Dissertation

[6] Author(s) First name or initials. Surname, "Title of thesis or dissertation," Qualification Type, Department Name, University Name, City of University, State/Country, Year.

Manual

[7] *Name of Manual/Handbook,* Xth ed., Name of Company/Institution, Location of Company (e.g. City, State, Country), year of publication, pp. xxx-xxx (page numbers if relevant). Accessed: Data. [Online]. Available: URL

10. Internet Sources

[8] Author initials. Last name. "Page title." Website Name. URL (accessed Month Day, Year).

11. Patents

[9] Author(s) First name or initials. Surname, "Title of Patent," Country of Patent + Patent Number, Date of Issue.

- **Research Colloquium (ACM/IEEE Format)**

  The students are required to submit a 5-10 pager-research report in IEEE journal article format.

COVER/TITLE PAGE (Thesis/Capstone)

|

(9 spaces)

(TITLE)

(6 spaces)

Student's Name1
Student's Name 2
Student's Name 3

(6 spaces)

A Thesis/Capstone Presented to the
Faculty of Department of Computing and Informatics
City College of Calamba
Calamba City

(6 spaces)

In Partial Fulfillment
Of the Requirements or the Degree of
Bachelor Science in Computer Science

(6 spaces)

Date

Prepared By:

Dr. Jacqueline A. Dela Torre
Research Facilitator, Capstone 1 and Capstone 2

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Research Facilitator, Thesis 1 and Thesis 2

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